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Novel mathematical modeling, performance analysis, and design charts for the typical hybrid photovoltaic/phase-change material (PV/PCM) system

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HIGHLIGHTS

- Novel PV/PCM model is introduced with better accuracy, computation time, and simplicity.
- Two innovative formulas for the PCM overall heat transfer coefficient are implemented.
- Computation time of the new PV/PCM model is less than 14 s with no mesh patterns.
- Melting time-interval is linear with PCM thickness, being 8.28 min/mm as a case study.
- Thermal efficiency of the system is rising by larger rates at lower solar radiation levels.

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ABSTRACT

As a type of hybrid PV/T solar collector, the PV/PCM system is typically used to produce and store both electrical and thermal energy in a variety of applications. Specialized computational-fluid-dynamic (CFD) models such as Ansys FLUENT and COMSOL are currently used to simulate the operation of these systems. However, the relatively long computation time of these models, which can range from a few hours to a few days for even simple system variables, poses a challenge. Therefore, the effort in this study has been devoted to developing an alternative mathematical model for PV/PCM systems that is sufficiently accurate, fast to compute, and simple to use. The proposed model is based on an exact five-parameter photovoltaic model coupled with a well-established photovoltaic/thermal network. The overall heat transfer coefficient of the thermal component (PCM) within the system's network is determined using two novel formulas in relation to the PCM's average temperature across its solid, melting, and liquid phases. The proposed model was implemented in MATLAB as a code that utilizes straightforward numerical functions without the use of mesh patterns. The model was then validated via both CFD and experimental verification. The results indicated that the model is reasonably accurate in terms of percentage errors and correlation coefficients. The model's computation time for system temperatures and PV power is less than 14 s, with a total simulation time of 2.2 h at a time step of 2.5 min, compared to 16 h for CFD. Thus, the developed model can substantially help in conducting comprehensive investigations of PV/PCM performance characteristics, parametric analyses, and design charts using a diverse range of system variables. Two performance indexes for the PV/PCM system were introduced: the PCM melting time interval and thermal efficiency. It was found that the PCM performs better in its upper layer, and this layer, which accounts for only about 5%–6% of the total PCM thickness, primarily governs the aforementioned performance indexes as well as the overall heat transfer coefficients. The melting time interval was found to increase linearly with PCM thickness, whereas the thermal efficiency increases logarithmically with incident solar radiation, with greater rates at lower solar radiation levels. However, over the three phases, the PV efficiency varies within a narrow range of 15.5–17.3%. The parametric analyses showed that thermal efficiency increases approximately fivefold as ambient temperature increases from 0 to 35 °C, but decreases by 55% as wind speed increases from 0 to 4 m/s.

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Nomenclature

A	surface area (m^2)
C	temperature coefficient of band-gap energy (eV/K)
C_p	specific heat capacity (J/(kg K))
E	absorbed energy per unit area (J/m^2)
E_e	rate of electrical energy available per unit area (W/m^2)
E_g	silicon band-gap energy (eV)
e	normalized root mean square of error (%)
F	view factor of PV panel
G	incident solar radiation (W/m^2)
h_d	conductive heat transfer coefficient (W/(m^2 K))
h_r	radiative heat transfer coefficient (W/(m^2 K))
h_v	convective heat transfer coefficient (W/(m^2 K))
h_{latent}	latent heat (J/kg)
I	current (A)
K	thermal conductivity (W/(m K))
k	Boltzmann's constant (J/K)
N_s	number of PV cells in series
P	output power (W)
PCM	phase change material
PV/T	photovoltaic/thermal
q	heat flux (W/m^2)
r	linear coefficient of correlation
S	rate of total energy absorbed per unit area of PV cell (W/m^2)
T	average temperature (K or °C)
t	time (s)
U	overall heat transfer coefficient (W/(m^2 K))
U_f	overall front loss coefficient from PV cell to ambient (W/(m^2 K))
U_b	overall back loss coefficient from PCM lower layer to ambient (W/(m^2 K))
U_L	total overall heat loss coefficient (W/(m^2 K))
V	voltage (V)
V_w	wind speed (m/s)
z	thickness (m)

Greek symbols

α	absorptivity
β	tilt angle (degree)
γ	rate of convection rise (K^{-1})
ε	emissivity
η	efficiency (%)
λ	position parameter
μ_{Isc}	temperature coefficient of short-circuit current (A/K)
μ_{Voc}	temperature coefficient of open-circuit voltage (V/K)
ρ	density (kg/m^3)
σ	Stefan-Boltzmann's constant (W/(m^2 K^4))
τ	transmissivity
ΔR_v	magnitude of convection rise (W/(m K))
Δh	change in enthalpy (J/kg)

Subscripts

a	ambient
Al	aluminum
back	back surface of system
bk	backside
c	PV cell
ch	aluminum chamber
ch1	upper side of aluminum chamber
ch2	lower side of aluminum chamber
front	front surface of system
g	glass
m	PCM melting
mp	maximum power point
oc	open circuit
ov	overall
p	PV backside
pcm	PCM
pcm1	PCM upper layer
pcm2	PCM lower layer
pv	PV panel
pv1	PV upper layers
pv2	PV lower layers
r	radiative
ref	reference condition
sc	short circuit
ted	tedlar
th	thermal
u	useful

1. Introduction

Solar energy systems with hybrid components, such as PV/T collectors, for generating multiple simultaneous energy outputs have superior overall performance as compared to systems with individual energy outputs. The major advantage of the PV/T collector is the delivery of both electrical and thermal energy within the same unit for different applications such as power generation, drying process, hot-water supply, water desalination, and space cooling and heating [1–8]. At the same time, the cooling effect of the thermal component can invoke better electrical performance of the PV component through thermal regulation action. A PV/T collector consists of an electrical component, photovoltaic (PV) panel, and one or more thermal components for heat extraction. In recent literature, various active and passive techniques for the thermal component were used with various numerical simulations and experimental investigations, including modeling and performance analysis of flat-plate and concentrating PV/T collectors [9–15]. In fact, active techniques can enhance the heat release process, resulting in higher overall system performance than it might be possible with passive techniques. However, power requirements and maintenance costs in general could be greater. In all cases, utilization of the electrical and thermal energies plays an important role in the cost-effectiveness of the PV/T system.

Numerous studies related to PV/T collectors of active thermal components are found in the literature. Herrando et al. [4] estimated the total annual performance of a glazed PV/T system for the UK domestic sector. It was found that 51% and 36% of electric power and hot water demands respectively can be covered by the system. Tiwari et al. [10] developed a thermal model for water flat-plate collectors of semi-transparent PV panels integrated with thermoelectric modules and a tube-in-sheet configuration. The experimental validations of the developed model at different climate conditions showed that root mean square deviations ranged from 3.45 to 27.67 % with correlation coefficients of 0.65 to 0.99. Hossain et al. [12] proposed a novel thermal absorber design of PV/T collector for experimental investigations at various mass flow rates. At various flow rates, the highest electrical and thermal efficiencies were 10.46 and 74.62 %, respectively. Yazdanifard et al. [16] simulated the performance of a tube PV/T collector for various design and operating conditions. The corresponding results showed that the maximum thermal efficiency was 65.4% in the laminar flow regime and 69.5% in the turbulent flow regime. Boumaaraf et al. [17] presented a comparison study between classical PV panel and water

PV/T collector using MATLAB numerical simulation. It was observed that the corresponding electrical and thermal efficiency reached 7% and 0%, and 6.26% and 57.72% respectively. Amanlou et al. [18] investigated the effects of applying a linear Fresnel concentrator and different diffuser designs on the PV temperature distribution of the PV/T solar collector using computational fluid dynamics (CFD). The electrical and thermal performance of the corresponding low concentrating PV/T collector was evaluated by experimental measurements. The results showed the new design improved the electrical and thermal efficiency of the PV/T collector by 36% and 42.2% respectively. Amori and Al-Najjar [19] derived a thermo-electrical model for a single-pass hybrid PV/T collector with an experimental validation for PV temperature of 1.91% error and 0.990 correlation coefficient. The model was applied to two case studies in Iraq with electrical and thermal efficiencies of 12.3% and 19.4%, and 9% and 22.8% for winter and summer, respectively.

As an effective thermal management system for the PV/T collector, phase change materials (PCMs) have lately gained widespread interest as a passive heat removal approach for (PV/T) collectors. Such an arrangement is specifically called PV/PCM system, where the PCM absorbs/rejects the latent heat during liquification/solidification phase over a narrower transition-temperature range and acts as a thermal battery. The various thermal and electrical characteristics of the PV/PCM collector are to be determined according to the design parameters and the operating conditions using numerical methods. Appropriate facilities are to be employed to transport the thermal energy from the PCM for final energy use.

Various mathematical models for the PV/PCM system have been developed and used in the literature, such as the CFD model [20–23], thermal resistance model [24], and finite element method [25]. Comprehensive numerical simulations of different system parameters and operating conditions are needed for profound performance analysis and optimum design of the PV/PCM system. However, only limited case studies were found in the literature due to limitations in the existing mathematical models and numerical methods [26]. One major reason for such problems is the quite long computation time required to conduct one simulation of a single design parameter or operating condition as a result of the convective heat transfer process within the PCM and the fine mesh size requirement of the numerical solution. Thus, it would be nearly impossible to carry out substantial number of simulations per one study. As can be found in the literature, various PCM-based solar collectors, including PV/PCM systems, have been theoretically and experimentally studied. Ma et al. [27] presented a detailed literature review related to PV/T technology using natural air circulation and ventilation, heat pipe, water cooling, heat pump, and PCM. The various aspects of PV/PCM systems were introduced and discussed. The paper demonstrated that a simple PV control module may not be economically viable if used exclusively for PV thermal regulation. As such, the PV/PCM system integrated with a solar thermal (ST) was investigated as an innovative technology for its potential to provide heating services in buildings. Khanna et al. [28] carried out various performance analyses to study the different effects of tilt angle, wind velocity and direction, ambient temperature, and PCM melting temperature on the melting process, PV temperature, heat extraction, and PV efficiency of the PV/PCM system. Savvakis and Tsoutsos [29] designed and manufactured a PV/PCM for the experimental evaluation of PV energy performance. The results of the study indicated that the PV/PCM operating temperature would be decreased by up to 26.6 °C, so the yearly power generation of the new system would increase by 5.7%. Mahmood [30] conducted an experimental study of the thermal energy storage characteristics of a double-pass solar air heater (SAH) integrated with paraffin wax PCM at climatic conditions of Baghdad city, Iraq. The experimental results showed that the temperature of the outlet air was enhanced over the ambient temperature by (1.5–6.5) °C after sunset for 5 h period and the daily efficiency of the SAH/PCM was 56%. Mahdi et al. [31] proposed an innovative configuration of PV/PCM-metal foam assembly for PV thermal management. The new configuration enables faster heat dissipation and energy recovery during PCM charging and discharging. The results indicated that by including an exterior foam layer, the PCM melting time and corresponding PV thermal management duration could be increased by up to 55% for 20-mm PCM with only a 5-mm thick metal-foam layer. In another study, Mahdi et al. [32] developed a new approach for PV cooling. It depends on employing a group of PCMs that are selected so that their thermo-physical properties rise gradually and in a pattern that provides a higher storage density for the cooling system. This also enables the system to retain greater energy content, which in turn removes the heat generated by the photovoltaic panels and allows for lower melting rates of the PCM and longer thermal management of the PV module. Sopian et al. [33] presented experimental investigations of four different PV/T and PV designs: water PV/T, water PV/T with PCM tank, nanofluid PV/T with nano-PCM tank, and conventional PV panel. The corresponding peak overall efficiency of the three PV/T systems were observed to be 47%, 63% and 85.7% respectively. Thus, all PV/T systems exhibited better performance than the conventional PV panel with the highest efficiency for the nanofluid/nano-PCM PV/T system. Li et al. [34] designed, fabricated and tested PV/PCM and PV/PCM-T systems under outdoor conditions for performance comparison with a single PV panel. The PV temperature difference reached up to 23 °C and the total output energy of the PV/PCM and PV/PCM-T systems increased by 5.2% and 74.3% respectively in respect to the single PV panel. Ma et al. [26] developed an improved 1-D thermal model for the PV/PCM system that incorporates the PCM convective heat transfer effect with less computation time. The new model was simulated in MATLAB with a mesh size of 1 mm and validated via the CFD model. The results indicated that the accuracy of the proposed model is of 0.5% normalized error and of computation time 300 times faster than the corresponding CFD model.

It is widely accepted that the commonly used CFD models for PV/PCM simulation, such as Ansys FLUENT and COMSOL, suffer from inherently long computation time when searching for an optimum system design over a wide range of system variables. To address this issue, this study proposes a modified PV/T mathematical model for the PV/PCM simulation. The convective heat transfer coefficients are evaluated using innovative formulas for the three zones of phase transition: solid, mushy, and liquid. The proposed model can serve the following purposes: (a) eliminating the need for mesh patterns, which would significantly reduce computation time; (b) solving the energy balance equations of the new PV/PCM thermal network model using simple built-in numerical functions within the MATLAB environment; (c) enabling optimum design and comprehensive investigations of system characteristics with sufficiently reasonable accuracy, computation time, and simplicity; and (d) constructing design charts for the performance validations that were carried out against both CFD and experimental results. The proposed model can precisely predict the thermal and electrical characteristics of the system, including the time profiles of network temperatures, heat fluxes, and heat transfer and loss coefficients. Also, the model can predict the time profiles for the maximum PV power with the electrical, thermal, and overall efficiencies of the system in addition to the time interval for PCM melting. To the best knowledge of the authors, no previous study has been found in the literature for developing and applying such a straightforward PV/PCM thermal network model with no mesh patterns.

2. PV/PCM system layout and mathematical modeling

2.1. PV/PCM system layout

A typical PV/PCM system is to be studied in the present research with a crystalline silicon PV panel as the electrical component and a rectangular aluminum chamber filled with PCM as the thermal component of the system, as shown in Fig. 1. The aluminum chamber is assumed to be in perfect thermal contact with the PV panel. RT35 is adopted in this work as the PCM since it is commercially available and widely used in the literature [35–38]. The other advantages of RT35 are the high latent

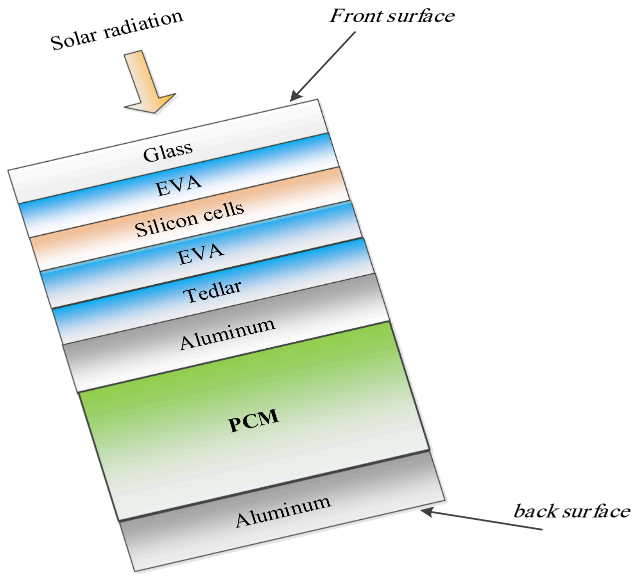

Fig. 1. Sectional view of the typical PV/PCM system.

heat of fusion (160 kJ/kg) and the phase-change temperature range that is compatible with PV thermal management application (29–36 °C) [39]. The basic properties of the different layers of the PV/PCM system: glass, EVA (Ethylene Vinyl Acetate), silicon cells, tedlar, and the chamber are summarized in Table 1. The specifications of the PV panel used in the present study at standard test conditions (STC) are listed in Table 2. Additional components might be engaged in the layout of the PV/PCM system to extract the generated heat for useful consumption.

2.2. PV/PCM mathematical modeling

As the typical PV/PCM system consists of two attached PV-electrical and thermal components, the system could be considered as a part of the hybrid PV/T collector family where both electrical and thermal energy of the collector are to be utilized in various related applications for better total system performance. Accordingly, the novel mathematical modeling of the PV/PCM system to be developed in the present paper will be based on a sound PV/T model with adequate modifications to achieve sufficiently reasonable accuracy, computation time, and simplicity. The chosen model for that purpose is taken from the previously published research of Amori and Al-Najjar [19] for thermo-electrical PV/T model. The paper proved that the developed model has the lowest percentage errors and the best correlation coefficients as compared to other related works. Following the methodology used for the chosen PV/T model [19], the rate of total energy absorbed per unit area of the PV cell (W/m^2) is found as:

$$S = (\tau_g \alpha_{pv}) G \tag{1}$$

This absorbed energy is divided into thermal energy and electrical energy as:

$$S = q_{th} + E_e \tag{2}$$

The rate of electrical energy available per unit area(W/m^2) is found as:

$$E_e = \eta_{pv} G \tag{3}$$

where $(\tau_g \alpha_{pv})$ is the product of transmittance and absorptance (assumed constant), G is the incident solar radiation (W/m^2), q_{th} is the rate of thermal energy absorbed per unit area (W/m^2) which is the thermal absorbed flux, and η_{pv} is the electrical efficiency of the PV panel. The detailed thermal and electrical models of the PV/PCM system are developed in the following two subsections.

Table 1
The basic properties of PV/PCM system layers [26].

Property	Glass	EVA	Silicon cells	Tedlar	Al chamber
Thermal conductivity (W/(m.K))	1.04	0.3	148	0.2	202
Density (kg/m^3)	2500	935	2330	1500	2791
Specific heat capacity (J/(kg.K))	750	2500	700	1090	871
Thickness (mm)	3.2	0.5	0.2	0.5	5
Transmissivity	0.96	—	—	—	—
Emissivity	0.88	—	—	—	0.1
Absorptivity	—	—	0.95	—	—

Table 2
PV panel specifications at STC [40].

Parameter	Value
Maximum output power	100 W
Voltage at maximum power point	17.7 V
Current at maximum power point	5.65 A
Open-circuit voltage	21.7 V
Short-circuit current	6.1 A
Temperature coefficient of open-circuit voltage	– 0.078 V/K
Temperature coefficient of short-circuit current	2 mA/K
Silicon band-gap energy	1.12 eV
Temperature coefficient of band-gap energy	– 2.677x10^{-4} eV/K
Surface area	0.585 m^2
Number of cells in series	36

2.2.1. Thermal model

According to the reference PV/T model of [19], the proposed thermal network for the typical PV/PCM system would be as shown in Fig. 2. This network has seven nodes of unknown temperatures. Each node temperature is assumed to be the average value over its cross-sectional area. From the energy balance of each node, then:

1- The temperature of front surface(glass) of the system.

$$T_g = \frac{h_{d,pv1} T_c + U_{front} T_a}{h_{d,pv1} + U_{front}} \tag{4}$$

where the conductive heat transfer coefficient of PV upper layers (EVA and glass) is.

$$h_{d,pv1} = (z_{EVA}/K_{EVA} + z_g/K_g)^{-1} \tag{5}$$

The overall heat loss coefficient from the front surface of the system to ambient due to convection and radiation is.

$$U_{front} = h_{v,front} + h_{r,front} \tag{6}$$

where the convective heat transfer coefficient of the front surface due to uniform wind speed V_w is [40].

$$h_{v,front} = 2.8 + 3V_w \tag{7}$$

This includes free and forced convection. The radiative heat transfer coefficient of the front surface with the sky is given by.

$$h_{r,front} = \sigma \varepsilon_g \frac{\left(T_g^4 - T_{sky}^4\right)}{(T_g - T_a)} F_{front,sky} \tag{8}$$

where the sky temperature can be found using [26]:

$$T_{sky} = 0.037536 T_a^{1.5} + 0.32 T_a \tag{9}$$

The view factor of the front surface with the sky in relation to the tilt angle β of the system is calculated as.

$$F_{front,sky} = \frac{1 + \cos(\beta)}{2} \tag{10}$$

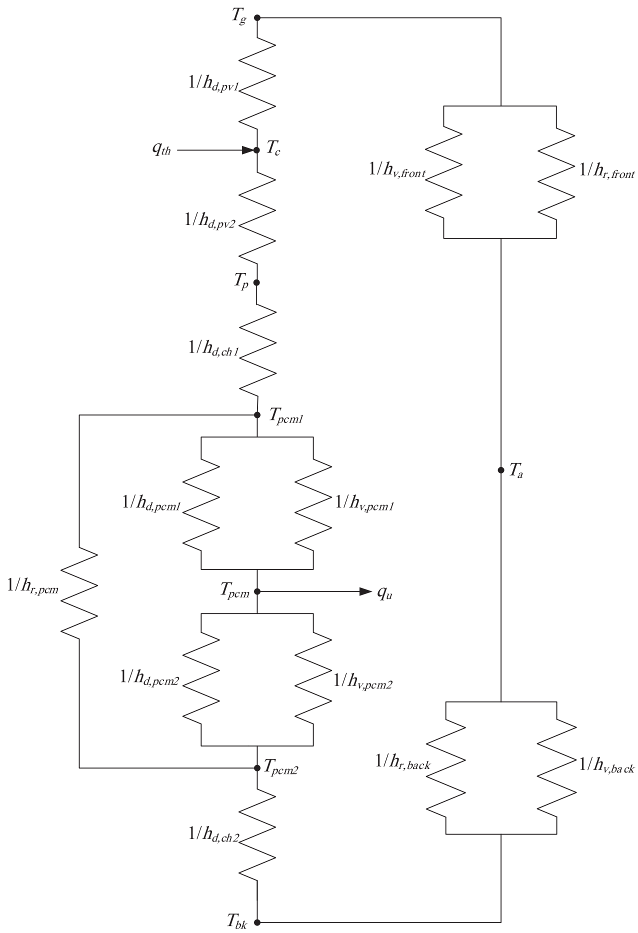

Fig. 2. Proposed thermal network for the typical PV/PCM system.

The corresponding front heat loss flux is calculated as.

$$q_{front} = U_{front}(T_g - T_a) \tag{11}$$

2- <u>The temperature of PV cell</u>.

$$T_c = \frac{(q_{th} + h_{d,pv1}T_g + h_{d,pv2}T_p)}{(h_{d,pv1} + h_{d,pv2})} \tag{12}$$

where the conductive heat transfer coefficient of PV lower layers (EVA and tedlar) is given as.

$$h_{d,pv2} = (z_{EVA}/K_{EVA} + z_{ted}/K_{ted})^{-1} \tag{13}$$

3- <u>The temperature of PV backside</u>.

$$T_p = \frac{h_{d,pv2}T_c + h_{d,ch1}T_{pcm1}}{h_{d,pv2} + h_{d,ch1}} \tag{14}$$

where the conductive heat transfer coefficient of the upper side of aluminum chamber is.

$$h_{d,ch1} = K_{Al}/z_{Al} \tag{15}$$

4- <u>The temperature of back surface of the system</u>.

$$T_{bk} = \frac{h_{d,ch2}T_{pcm2} + U_{back}T_a}{h_{d,ch2} + U_{back}} \tag{16}$$

where the conductive heat transfer coefficient of the lower side of aluminum chamber is.

$$h_{d,ch2} = h_{d,ch1} \tag{17}$$

The overall heat loss coefficient from the back surface of the system to ambient due to convection and radiation is.

$$U_{back} = h_{v,back} + h_{r,back} \tag{18}$$

The convective heat transfer coefficient of the back surface is assumed zero in this study due to space restrictions. The radiative heat transfer coefficient of the back surface with the ground is given by.

$$h_{r,back} = \sigma\varepsilon_{ch}(T_{bk} + T_{ground})(T_{bk}^2 + T_{ground}^2)F_{back,ground} \tag{19}$$

where the ground temperature is assumed to be equal to ambient temperature. The view factor of the back surface with the ground is found as.

$$F_{back,ground} = \frac{1 - \cos(\pi - \beta)}{2} \tag{20}$$

The corresponding back heat loss flux is calculated as.

$$q_{back} = U_{back}(T_{bk} - T_a) \tag{21}$$

The remaining three temperature nodes of Fig. 2, T_{pcm1}, T_{pcm2} and T_{pcm}, are related to the thermal component (PCM) of the system. As in the reference model of [19], the position of average PCM temperature T_{pcm} in the network divides the PCM into two layers: upper layer#1 and lower layer#2 with radiative heat transfer coefficient between T_{pcm1} and T_{pcm2}. Each PCM layer is assumed to have its own overall heat transfer coefficient including conduction and convection given as:

$$U_{pcm1} = h_{d,pcm1} + h_{v,pcm1} \tag{22a}$$

and.

$$U_{pcm2} = h_{d,pcm2} + h_{v,pcm2} \tag{22b}$$

where the conductive heat transfer coefficients are calculated by.

$$h_{d,pcm1} = K_{pcm}/z_{pcm1} \tag{23a}$$

and.

$$h_{d,pcm2} = K_{pcm}/z_{pcm2} \tag{23b}$$

where K_{pcm} is the thermal conductivity of the PCM. The thickness of each PCM layer, z_{pcm1} and z_{pcm2}, is to be found as.

$$z_{pcm1} = \lambda z_{pcm} \tag{24a}$$

and.

$$z_{pcm2} = (1 - \lambda)z_{pcm} \tag{24b}$$

where z_{pcm} is the total thickness of the PCM used in the system. The value of the parameter λ is to be determined by the actual position of the average PCM temperature according to the developed CFD numerical model in section 4.1. It is worth to be noted that for typical air and water PV/T collectors, the position of the average fluid temperature is somewhat beyond the middle of the fluid flow path within the collector ($\lambda > 0.5$).

The main challenge in the present research is the modeling of the convective heat transfer coefficients $h_{v,pcm1}$ and $h_{v,pcm2}$ in Eqns. (22a and 22b) with simple mathematical functions for the three PCM phases. For that purpose, it is proposed that, although at the solid phase the only heat transfer mechanism is by conduction, which is usually relatively small, as PCM starts melting, the heat transfer within the PCM increases as a function of the liquid fraction due to convection mechanism. Then, in the melted part of PCM, heat transfer attained saturation as liquid fraction is being unchanged. In this study, the liquid fraction within the PCM is assumed to be related to the average PCM temperature. Accordingly, new improved mathematical functions are developed from that of Ma et al. [26] for the convective heat transfer coefficients of PCM layer#1 and layer#2 as follows:

$$h_{v,pcm1} = \frac{\frac{\Delta R_{v,pcm}}{1+e^{-\gamma(T_{pcm}-T_m)}}}{z_{pcm1}} \tag{25a}$$

and.

$$h_{v,pcm2} = \frac{\frac{\Delta R_{v,pcm}}{1+e^{-\gamma(T_{pcm}-T_m)}}}{z_{pcm2}} \tag{25b}$$

where the average PCM melting temperature in terms of the solidus and liquidus temperatures is given by.

$$T_m = \frac{T_{solid} + T_{liquid}}{2} \tag{26}$$

The magnitude and the rate of rise of the PCM convection heat transfer in Eqns. (25) can be adjusted by the parameters $\Delta R_{v,pcm}$(W/(m. K)) and γ(K^{-1}) respectively with $\gamma \geq 1$ using the developed CFD model of the PV/PCM system as will be explained in section 4.1. Then, the overall heat transfer coefficients of Eqns. (22) for PCM layer#1 and layer#2 are to be evaluated in this paper using the two innovative formulas (27a and 27b) which are continuous functions of average PCM temperature (T_{pcm}) over the solid, melting, and liquid phases as:

$$U_{pcm1} = \frac{K_{pcm} + \frac{\Delta R_{v,pcm}}{1+e^{-\gamma(T_{pcm}-T_m)}}}{z_{pcm1}} \tag{27a}$$

and.

$$U_{pcm2} = \frac{K_{pcm} + \frac{\Delta R_{v,pcm}}{1+e^{-\gamma(T_{pcm}-T_m)}}}{z_{pcm2}} \tag{27b}$$

The heat fluxes in PCM layer#1 and layer#2 are given by.

$$q_{pcm1} = U_{pcm1}(T_{pcm1} - T_{pcm}) \tag{28a}$$

and.

$$q_{pcm2} = U_{pcm2}(T_{pcm} - T_{pcm2}) \tag{28b}$$

Applying the energy balance for the three nodes of PCM temperatures, then:

5- <u>The temperature of upper PCM layer</u>.

$$T_{pcm1} = \frac{h_{d,ch1}T_p + h_{r,pcm}T_{pcm2} + U_{pcm1}T_{pcm}}{h_{d,ch1} + h_{r,pcm} + U_{pcm1}} \tag{29}$$

The radiative heat transfer coefficient through the PCM (between T_{pcm1} and T_{pcm2}) is given by:

$$h_{r,pcm} = \sigma\frac{(T_{pcm1} + T_{pcm2})(T_{pcm1}^2 + T_{pcm2}^2)}{1/\varepsilon_1 + 1/\varepsilon_2 - 1} \tag{30}$$

where ε_1 and ε_2 are the emittances of the upper and lower sides of aluminum chamber respectively with $\varepsilon_1 = \varepsilon_2 = \varepsilon_{ch}$. The corresponding radiative heat flux is calculated as:

$$q_{r,pcm} = h_{r,pcm}(T_{pcm1} - T_{pcm2}) \tag{31}$$

6- <u>The temperature of lower PCM layer</u>.

$$T_{pcm2} = \frac{U_{pcm2}T_{pcm} + h_{r,pcm}T_{pcm1} + h_{d,ch2}T_{bk}}{U_{pcm2} + h_{r,pcm} + h_{d,ch2}} \tag{32}$$

7- <u>The average PCM temperature</u>.

The temperature of the last node, the average PCM temperature, is found using the PCM absorbed energy per unit area E_{pcm}(J/m^2). From Fig. 2, E_{pcm} at time t can be calculated as:

$$E_{pcm} = (q_{pcm1} - q_{pcm2})t \tag{33}$$

Then:

$$E_{pcm} = U_{pcm1}(T_{pcm1} - T_{pcm})t - U_{pcm2}(T_{pcm} - T_{pcm2})t \tag{34}$$

On the other hand, from the thermodynamic point of view, the change in enthalpy Δh (J/kg) of the PCM with the temperature variation ΔT is given by [41]:

$$\Delta h = (C_p)_{pcm}\Delta T \tag{35}$$

Hence:

$$E_{pcm} = z_{pcm}(\rho C_p)_{pcm}\Delta T \tag{36}$$

Applying Eqn. (36) for each of the solid, melting, and liquid phase, it can be found that:

$$E_{pcm} = \begin{cases} (T_{pcm} - T_a)z_{pcm}(\rho C_p)_{solid}, for T_{pcm} \leq T_{solid} \\ (T_{pcm} - T_{solid})z_{pcm}(\rho C_p)_{melting} + E_{sensible}, for T_{solid} < T_{pcm} < T_{liquid} \\ (T_{pcm} - T_{liquid})z_{pcm}(\rho C_p)_{liquid} + E_{sensible} + E_{latent}, for T_{pcm} \geq T_{liquid} \end{cases} \tag{37}$$

where the total sensible energy absorbed (J/m^2) during the solid phase is given by:

$$E_{sensible} = (T_{solid} - T_a)z_{pcm}(\rho C_p)_{solid} \tag{38}$$

and the total latent energy absorbed (J/m^2) during the melting phase is given by:

$$E_{latent} = (T_{liquid} - T_{solid})z_{pcm}(\rho C_p)_{melting} \tag{39}$$

Then, equating Eqns. (34 and 37) for each phase, and solving for the average PCM temperature:

$$T_{pcm} = \begin{cases} \frac{(U_{pcm1}T_{pcm1} + U_{pcm2}T_{pcm2})t + c1}{(U_{pcm1} + U_{pcm2})t + z_{pcm}(\rho C_p)_{solid}}, for T_{pcm} \leq T_{solid} \\ \frac{(U_{pcm1}T_{pcm1} + U_{pcm2}T_{pcm2})t + c2}{(U_{pcm1} + U_{pcm2})t + z_{pcm}(\rho C_p)_{melting}}, for T_{solid} < T_{pcm} < T_{liquid} \\ \frac{(U_{pcm1}T_{pcm1} + U_{pcm2}T_{pcm2})t + c3}{(U_{pcm1} + U_{pcm2})t + z_{pcm}(\rho C_p)_{liquid}}, for T_{pcm} \geq T_{liquid} \end{cases} \tag{40}$$

The constants $c1$,$c2$ and $c3$ are given by:

$$c1 = T_a z_{pcm}(\rho C_p)_{solid} \tag{41}$$

$$c2 = T_{solid}z_{pcm}(\rho C_p)_{melting} - E_{sensible} \tag{42}$$

$$c3 = T_{liquid}z_{pcm}(\rho C_p)_{liquid} - (E_{sensible} + E_{latent}) \tag{43}$$

The density ρ of the PCM during the melting phase is taken as the average of the solidus and liquidus densities, while the specific heat capacity during the melting phase $(C_p)_{melting}$ is being calculated in this study from the PCM latent heat h_{latent} as:

$$(C_p)_{melting} = h_{latent}/\Delta T_m \tag{44}$$

where the range of the PCM melting temperature is.

$$\Delta T_m = T_{liquid} - T_{solid} \tag{45}$$

It is to be noted that the PCM absorbed energy during any phase could be utilized to produce useful thermal energy, however it can only be stored during the melting phase of PCM. Table 3 presents the thermophysical properties for the three phases of the PCM RT35 used in the present study.

An important system thermal characteristic is the total overall heat loss coefficient U_L which can be calculated as in solar thermal collectors and neglecting side losses [40]:

$$U_L = U_f + U_b \tag{46}$$

In this case, the front and back loss coefficients in Eqn. (46) are to be evaluated as from PV cell temperature and lower PCM-layer temperature to ambient respectively, then:

Table 3
Thermo-physical properties of the PCM (Rubitherm-RT35) [39].

Property	Solid phase	Melting phase	Liquid phase
Operating temperature (°C)	≤ 29	$29 < T < 36$	≥ 36
Maximum operating temperature (°C)	65	65	65
Thermal conductivity (W/(m.K))	0.2	0.2	0.2
Density (kg/m^3)	860	815	770
Specific heat capacity (J/(kg.K))	2000	Eqn. (44)	2000
Latent heat (J/kg)	—	160,000	—

$$U_f = \left(1/h_{d,pv1} + 1/U_{front}\right)^{-1} \tag{47}$$

and.

$$U_b = \left(1/h_{d,ch2} + 1/U_{back}\right)^{-1} \tag{48}$$

Finally, the thermal efficiency of the PV/PCM system is introduced as:

$$\eta_{th} = \frac{q_u}{G} \tag{49}$$

The available useful heat flux q_u at the average PCM temperature can be calculated in terms of the heat fluxes in the upper and lower layers of the PCM as:

$$q_u = q_{pcm1} - q_{pcm2} \tag{50}$$

2.2.2. Electrical model

To calculate the rate of electrical energy available per unit area of the system E_e(W/m^2) in Eqns. (2 and 3), the PV efficiency η_{pv} needs to be evaluated based on the maximum electrical power generated by the PV panel at each time. For exact solution, the 5-parameter model of the PV equivalent circuit (Fig. 3) is adopted in the present study for that purpose [40].

The governing equations for the 5-parameter model are given as:

$$I_{pv} = I_L - I_D - I_{sh} \tag{51}$$

$$I_D = I_o.\left(e^{(V_{pv}+R_sI_{pv}/a)} - 1\right) \tag{52}$$

$$I_{sh} = \frac{V_{pv} + R_sI_{pv}}{R_{sh}} \tag{53}$$

The output power of the PV panel is calculated by:

$$P_{pv} = V_{pv}I_{pv} \tag{54}$$

where I_{pv} and V_{pv} are the output current and voltage of the PV panel respectively. The five characteristic parameters of the model are light current I_L, diode reverse saturation current I_o, modified ideality factor a, series resistance R_s, and shunt resistance R_{sh}. Accordingly, the maximum PV power and its corresponding efficiency will be calculated in three steps as follows.

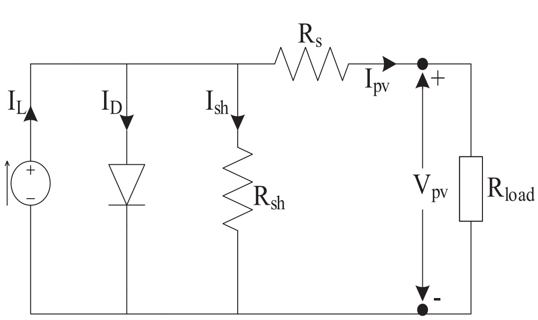

Fig. 3. Equivalent electrical circuit of the PV panel.

Step 1: Solving the 5-parameter model.

In this step, the five parameters of the PV model are to be found numerically at the reference condition (STC) of $G_{ref} = 1000$ W/m^2 and $T_{c,ref} = 25$°C using the following improved boundary conditions for better convergence and accuracy [19]:

1. at short circuit: $I_{pv} = I_{sc,ref}, V_{pv} = 0$.
2. at open circuit: $I_{pv} = 0, V_{pv} = V_{oc,ref}$.
3. at maximum power point:$I_{pv} = I_{mp,ref}, V_{pv} = V_{mp,ref}$.
4. at maximum power point: $\left.\frac{dP_{pv}}{dV_{pv}}\right|_{mp,ref} = 0$.
5. at open circuit: $\frac{\partial V_{oc}}{\partial T_c} = \mu_{V_{oc}}$.

Step 2: Calculating model parameters at the operating condition.

The model parameters from step 1 are to be translated to that at the operating condition of given incident solar radiation G and PV cell temperature T_c using the following relationships [40]:

$$a = a_{ref}T_c/T_{c,ref} \tag{55}$$

$$I_L = \left(G/G_{ref}\right)\left[I_{L,ref} + \mu_{I_{sc}}\left(T_c - T_{c,ref}\right)\right] \tag{56}$$

$$E_g = E_{g,ref}\left[1 + C\left(T_c - T_{c,ref}\right)\right] \tag{57}$$

$$I_o = I_{o,ref}\left(T_c/T_{c,ref}\right)^3 \times e^{\left[\left.\frac{E_g}{kT}\right|_{T_{c,ref}} - \left.\frac{E_g}{kT}\right|_{T_c}\right]} \tag{58}$$

$$R_{sh} = R_{sh,ref}G_{ref}/G \tag{59}$$

$$R_s = R_{s,ref} \tag{60}$$

where k is the Boltzmann's constant, E_g is the silicon band-gap energy, C is the temperature coefficient of band-gap energy, $\mu_{I_{sc}}$ is the temperature coefficient of short-circuit current, and $\mu_{V_{oc}}$ is the temperature coefficient of open-circuit voltage.

Step 3: Finding the maximum PV power and efficiency.

Using the model parameters of step 2, the PV power at the maximum power point (P_{mp}) is to be found numerically such that:

$$P_{mp} = P_{pv}\big|_{mp} \tag{61}$$

Then, the corresponding PV efficiency is calculated by:

$$\eta_{pv} = \frac{P_{mp}}{A_{pv}G} \tag{62}$$

where A_{pv} is the surface area of the PV panel. Finally, the total overall efficiency of the PV/PCM system is given by:

$$\eta_{ov} = \eta_{pv} + \eta_{th} \tag{63}$$

3. MATLAB simulation of the PV/PCM system

The developed PV/PCM model of section two is formulated as a computer program to be implemented in MATLAB [42] according to the flow chart of Fig. 4. As the thermal and electrical parts of the MATLAB model are strongly coupled and time-dependent, an iterative procedure with suitable initial values is adopted to meet the convergence conditions of acceptable errors. The set of seven nodal temperature equations (4, 12, 14, 16, 29, 32, and 40) of the thermal network of Fig. 2 are solved simultaneously. While the maximum power and its corresponding efficiency of the PV model are calculated in three steps using the 5-parameter equivalent circuit of Fig. 3 as described in subsection 2.2.2. The nodal temperatures and PV parameters are found numerically using built-in functions of the MATLAB. It can be noticed that the novel PV/PCM model developed in the present research does not need for any mesh patterns in contrast to currently used numerical models. PV/PCM system variables comprise several input and output data, where the input data for the computer program include the design parameters and

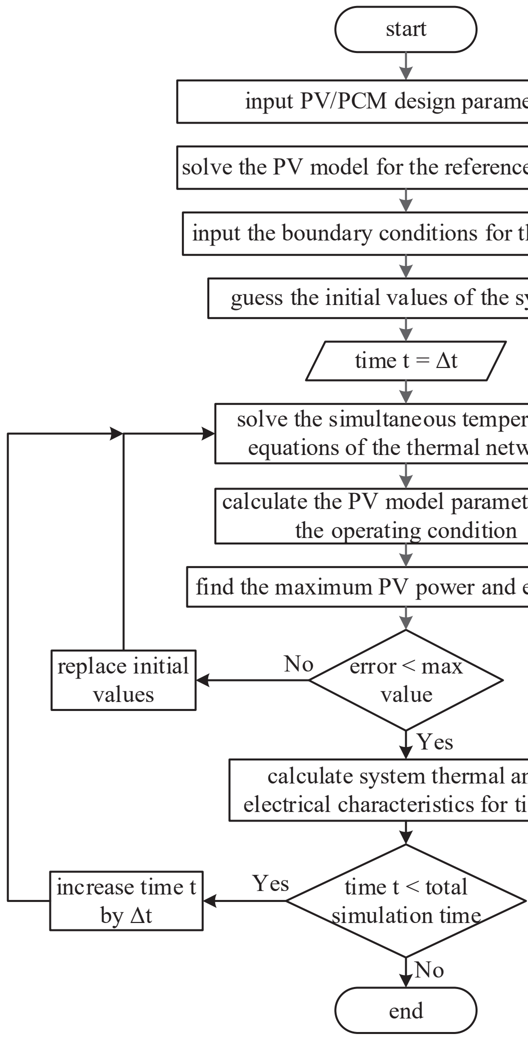

Fig. 4. Flow chart of the computer program.

boundary conditions of the system. In the present paper, the boundary conditions are taken as the incident solar radiation, ambient temperature, wind speed, tilt angle, and PCM thickness. The total simulation time and the time step for the program are to be specified according to the process length and the required resolution respectively. The performance of the PV/PCM system expressed by its thermal and electrical characteristics constitutes the output data of the program. It includes the time profiles for the seven node temperatures, the corresponding heat fluxes, and heat transfer and loss coefficients of the thermal network of Fig. 2. In addition, the time profile of the PCM absorbed energy is calculated. Also, the output comprises the time profiles for the maximum PV power with the electrical, thermal, and overall efficiencies of the system. The time interval for PCM melting is being found from the time profile of average PCM temperature. It worth to be noted that the parameters λ, $\Delta R_{v,pcm}$ and γ in the MATLAB model for the position of average PCM temperature, the magnitude, and the rate of rise of the PCM convection heat transfer respectively are to be adjusted using a developed CFD numerical model of the PV/PCM system as will be explained.

4. Numerical and experimental validations

The novel PV/PCM model presented in this research will be verified against two different approaches: a developed CFD simulation and published experimental results. For that purpose, a statistical validation is adopted to specify numerically the model accuracy in terms of the normalized root mean square error (e) [26] and the linear coefficient of correlation (r) [19] over the full range of data, as explained in the following subsections.

4.1. Numerical validation

As the CFD numerical solution is widely used for PCM-based systems. A CFD model is designed and constructed in the present study using Ansys FLUENT [43] for the purpose of both adjustment of the three parameters (position parameter λ, magnitude of rise $\Delta R_{v,pcm}$, and rate of rise γ) of MATLAB program and validation of the proposed model. The CFD model is used to predict the time profiles of PV cell, upper PCM layer and lower PCM layer temperatures. The average PCM temperature is calculated for selected times of the simulation. The same initial and boundary conditions, as well as governing equations and system's dimensions used in the MATLAB model, are applied to a fixed grid that defines the computation domain in Ansys FLUENT. The independence of numerical simulation with respect to timestep and mesh size is thoroughly examined in the preliminary CFD tests as shown in Fig. 5. The system with a grid size of 10200 and a timestep of 0.15 s are chosen considering both the accuracy and computation cost requirements. An additional grid refinement is determined to be non-significant in terms of the liquid fraction evolution. The CPU time for running the CFD cases is about 16 h on Intel (R)/Core (TM) i7-4810MQ.

It should be noted that according to the analysis of [26], the solidus temperature of PCM RT35 was found to be effectively at 32 °C. Thus, this value is adopted in the present research. The CFD simulation is carried out for the boundary conditions of incident solar radiation 1000 W/m^2, ambient temperature 25 °C, wind speed 0 m/s, tilt angle 90°, and PCM thickness 30 mm. Such set of boundary conditions was chosen in order to get a sufficient total simulation time that includes the complete profile for the solid, melting, and liquid phases of the PCM with a moderate computational time of around 20 h. At time 30 min, when the maximum difference between the temperatures of upper and lower PCM layers is noticed, the three node temperatures of the PCM (Fig. 2) with their corresponding positions is shown in Fig. 6 according to the results of the CFD simulation. It is observed from Fig. 6 that the position of average PCM temperature is being at 1.7 mm from the upper side of the PCM. Thus, the position parameter λ is set at 5.7% according to Eqn.(24a). On the other hand, the value of average PCM temperature is found to be very closer to the lower layer temperature than to the upper layer temperature of the PCM as displayed in Fig. 6. The developed CFD model is also used to calibrate the PCM convection model of Eqns. (25) to find the optimum values of the magnitude of rise $\Delta R_{v,pcm}$ and the rate of rise γ for best numerical validation results i.e., minimum errors and maximum correlation coefficients by fine tuning. The final values of $\Delta R_{v,pcm}$ and γ are found to be 8.74 W/(m.K) and 1.12 K^{-1} respectively. It should be noted that the values of the parameters λ, $\Delta R_{v,pcm}$ and γ in the MATLAB model are being unchanged throughout this research irrespective of the boundary conditions of the PV/PCM system.

The time profiles of PV cell, upper PCM layer and lower PCM layer temperatures by the CFD model along with those by the present MATLAB model are shown in Figs. 7–9 respectively. It can be seen from Figs. 7 and 8 that the temperature predictions based on the present model are consistent with those made by the CFD model beyond the early stage of PCM melting (>40 min). During this stage, the CFD model exhibited slower response with rather higher predicted temperature in respect to the present model. This is due to the fact that the developed CFD model utilized a simplified single-layer representation of the PV panel with average equivalent thermophysical properties of the actual PV layers (glass, EVA, silicon cells and tedlar). Note that the melting phase persisted along (10–100) minutes. On the other hand, Fig. 9 shows very good agreement for all range of data as the PCM is well-simulated in both CFD and present models. Applying the statistical validation between the two models for each of the three temperature profiles, the

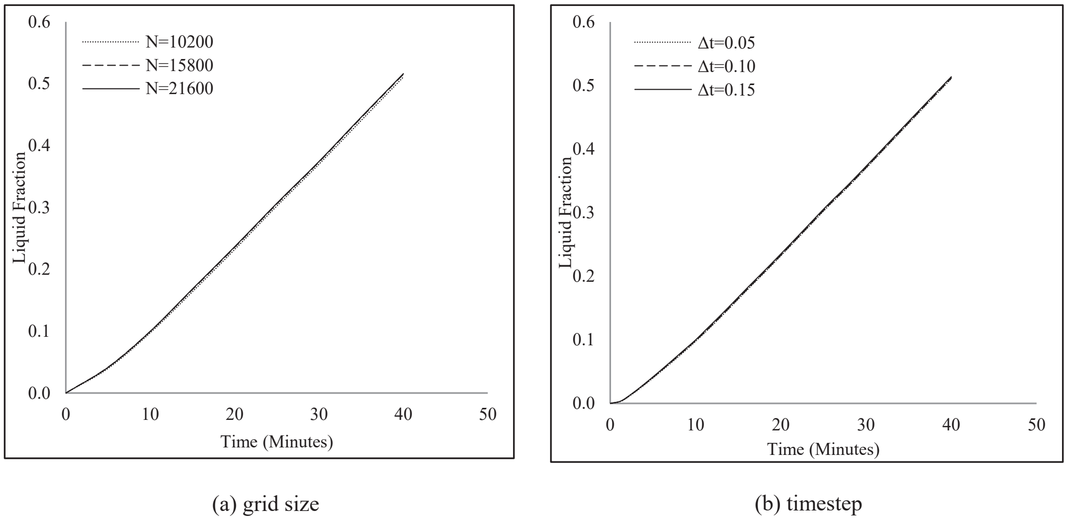

Fig. 5. The grid size and timestep effects on the PCM liquid-fraction evolution during the melting duration.

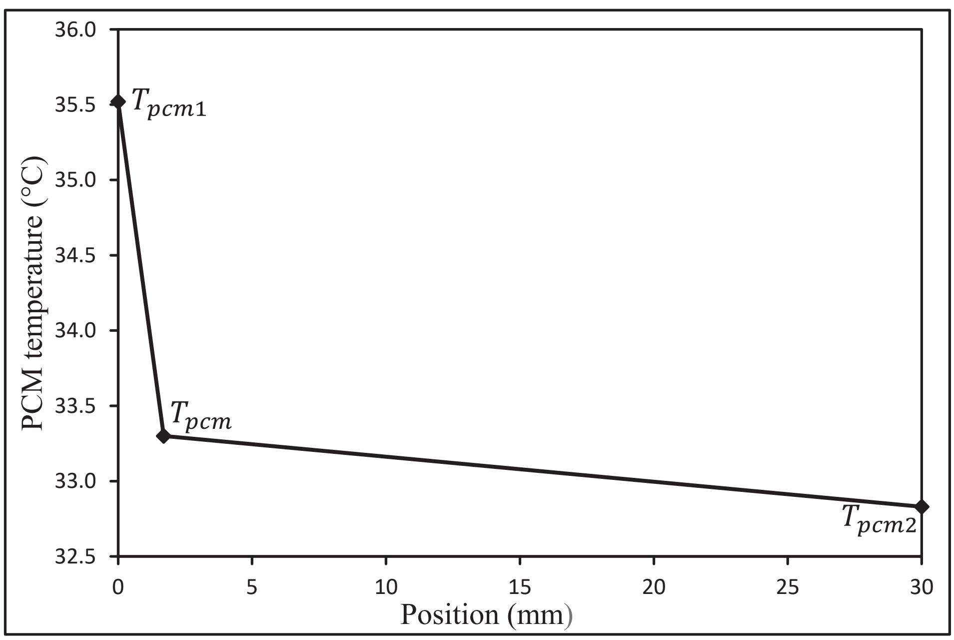

Fig. 6. PCM temperature with position at simulation time 30 min.

percentage errors are found to be 2.5%, 2.4% and 1.4% respectively. While the correlation coefficients are 0.987, 0.988 and 0.997 respectively which indicates good accuracy of the proposed PV/PCM MATLAB model. It is worth to be noted that beyond 30 min, the temperature profiles of upper and lower PCM layers (Figs. 8 and 9) for each CFD and present model are getting closer to each other such that all PCM will be of identical temperatures.

4.2. Experimental validations

Two experimental results from previously published works on typical PV/PCM systems are chosen for further validation of the present proposed model. The same boundary conditions and design parameters of the PV/PCM system in the two reference studies are applied in the present validations. These include heat transfer coefficients of 10 and 5 W/(m^2 K) on the front and rear surfaces of the PV/PCM system, inward solar radiation of 1000 W/m^2, and an initial PCM temperature and ambient air temperature of 20 °C. The first experimental work is that by Biwole et al. [44]. Fig. 10 shows the time profile for the PV cell temperature in Ref. [44] along with that by the present model. The statistical validation gave a percentage error of 1.4% and a correlation coefficient 0.999. Thus, the present model has a very good agreement with the experimental results of Ref. [44]. The second selected experimental work is by Huang et al. [45]. The corresponding time profiles for the PV cell temperature as shown in Fig. 11 are found to be very close with a percentage error of 1.3% and a correlation coefficient 0.998. Accordingly, the proposed PV/PCM model shows a high experimental validity against both reference studies [44] and [45].

By comparing the CFD validation (Fig. 7) with the experimental validations (Fig. 10 and Fig. 11) for the PV cell temperature, it can be indicated that the present proposed model displays an excellent match with the two experimental results for the whole range of data, whereas it shows some noticeable mismatch with the CFD result during the early stage of PCM melting as explained above. Nevertheless, it can be concluded that the excellent accuracy indicated by Table 4 confirms the validity of the proposed PV/PCM model. Besides, it shows that the

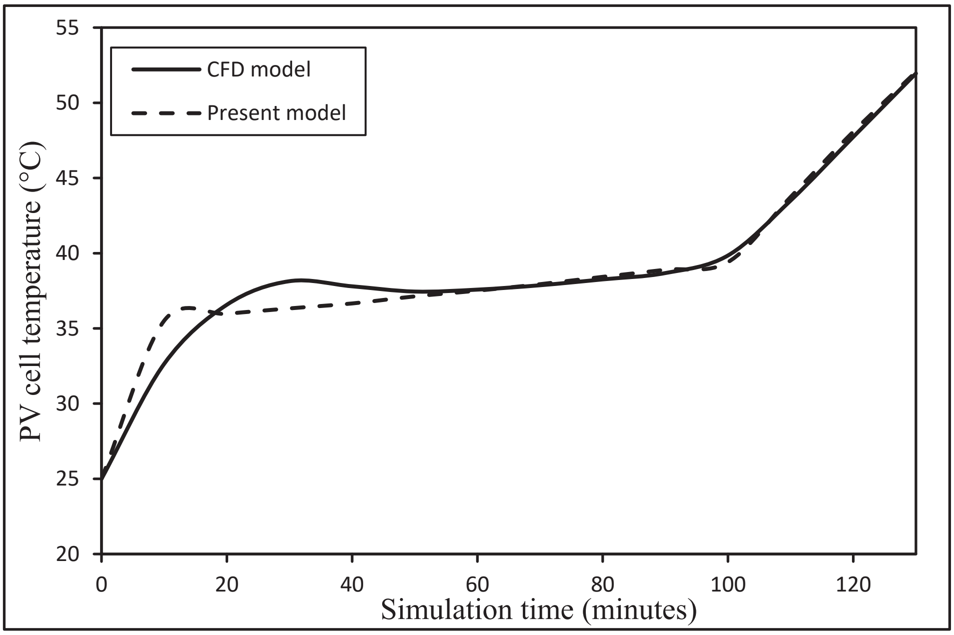

Fig. 7. Time profiles of PV cell temperature by the CFD model and the present proposed model.

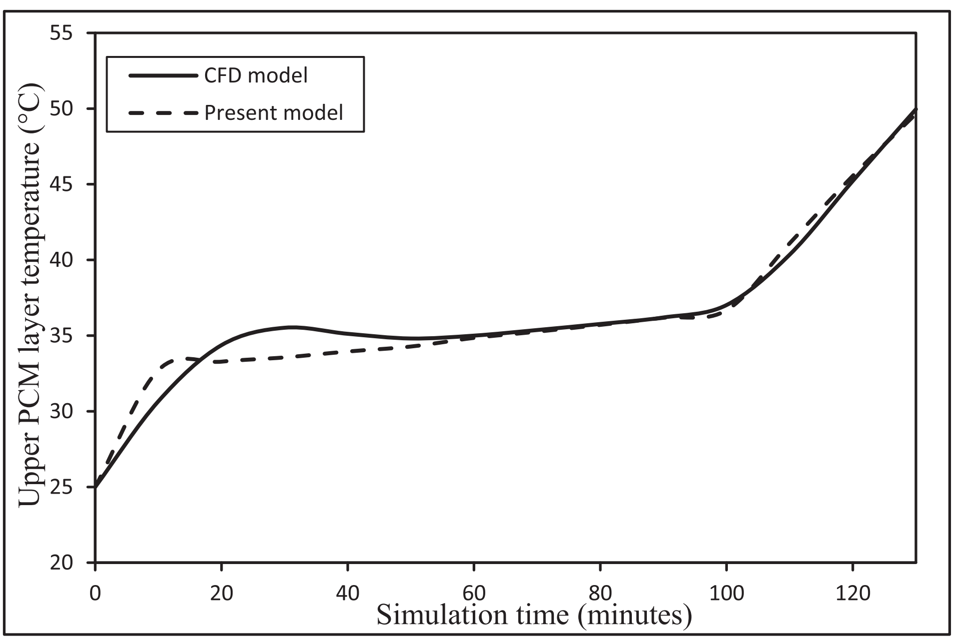

Fig. 8. Time profiles of upper PCM layer temperature by the CFD model and the present proposed model.

present model is closer to the experimental results than to the CFD results, which offers a practical bias for the proposed model. In fact, as the three validation cases represent various PV/PCM systems of different design parameters and boundary conditions, they all prove the reliability of the present model, which is necessary for further applications.

5. Performance analysis of the PV/PCM system

As shown in previous sections, the novel PV/PCM model presented in this paper is found to be sufficiently reliable and of simple numerical algorithm of solution that eliminates the use of mesh patterns in the simulation. Moreover, the computation time of the corresponding MATLAB program using the boundary conditions for CFD validation of section 4.1 is less than 14 s for total simulation time 2.2 h at time step 2.5 min. Thus, the novel model greatly shortens the computation time by over 4000 times in respect to the CFD model used in this paper. The present model is adopted for performing comprehensive investigations of the system involving 4800–5300 simulation runs at a wide range of system variables. The performance analysis of the PV/PCM system is presented and discussed in this section in terms of two sets of data for a case study of boundary conditions: incident solar radiation 650 W/m^2, ambient temperature 20 °C, wind speed 1 m/s, tilt angle 45°, and PCM thickness 25 mm. The first set is the timewise variations of system characteristics over the three phases of the PCM, while the second set is the parametric study of system efficiencies (PV, thermal and overall) and the PCM melting time-interval as explained in the following two subsections.

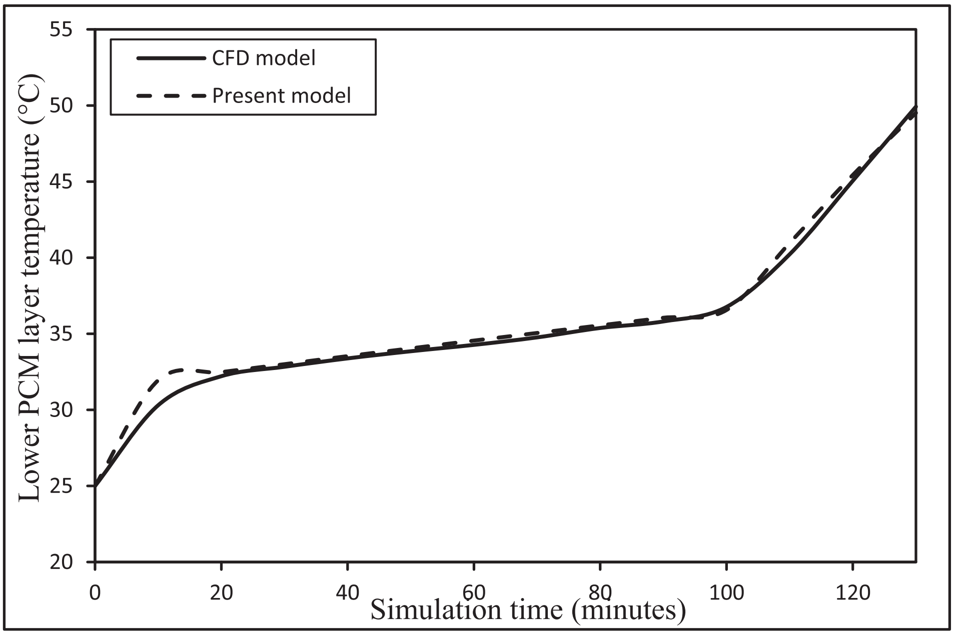

Fig. 9. Time profiles of lower PCM layer temperature by the CFD model and the present proposed model.

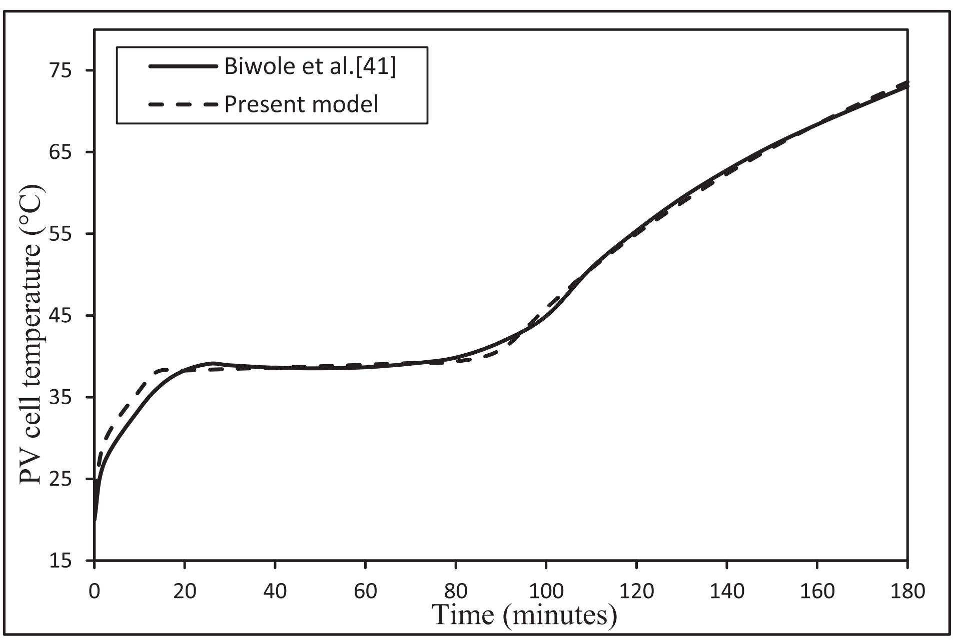

Fig. 10. Time profiles of PV cell temperature by Biwole et al. [44] and the present proposed model.

5.1. Time profiles of system characteristics

PV/PCM system characteristics for the designated case study are illustrated by Figs. 12–17 with total simulation time of 7 h. Fig. 12 shows the main system temperatures: PV cell (T_c) and the PCM (layer#1-T_{pcm1}, layer#2-T_{pcm2} and the average-T_{pcm}). It is noticed that temperatures' profiles are characterized by three different time zones according to PCM average temperature. The solid phase is of sharp rise from ambient (20 °C) to solidus temperature (32 °C) for 0–30 min due to low heat absorbing rate. While during melting phase, 30–240 min, a narrow transition temperature of 32 °C to 36 °C is encountered as large heat is being stored within the PCM. For the third time zone, the liquid phase is started at 240 min when PCM temperature reaches the liquidus temperature of 36 °C. In this zone, a moderate temperature rise is observed where the heat transfer rate attained a steady value after PCM melting. The three PCM temperatures of layer#1, layer#2, and the average are found to be very close over the simulation time as the operating heat flux is relatively low due to somewhat low solar radiation with high heat loss to ambient caused by the wind speed. The melting time interval, which is the maximum available time for thermal energy storage, is 3.5 h. The PV cell temperature (T_c) is within 1.7 °C higher than the PCM, while the glass temperature (T_g), although not shown for convenience, is within 1.0 °C lower than the PV cell during the melting phase.

The overall heat loss coefficients from the system to ambient (U_f, U_b and U_L) are shown in Fig. 13. It can be noticed that the profile of front loss coefficient (U_f) is somewhat decreasing over the whole simulation time however with larger rate during the solid phase as PV cell temperature is increasing (Fig. 12). The value of front loss is being relatively

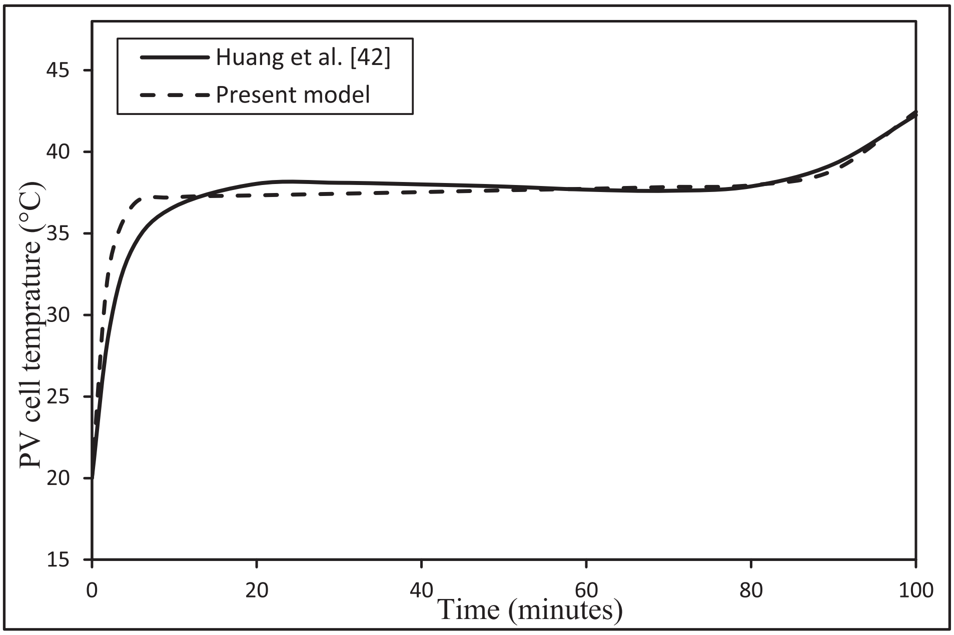

Fig. 11. Time profiles of PV cell temperature by Huang et al. [45] and the present proposed model.

Table 4
Statistical results of the numerical and experimental validations.

Validation	T_c		T_{pcm1}		T_{pcm2}	
	e%	*r*	*e%*	*r*	*e%*	*r*
CFD	2.5	0.987	2.4	0.988	1.4	0.997
Experimental:						
Biwole et al.[44]	1.4	0.999	N/A	N/A	N/A	N/A
Huang et al. [45]	1.3	0.998	N/A	N/A	N/A	N/A

high due to the considerable effect of the large deviation of sky temperature from ambient (>10 °C) at low wind speed. During the melting phase the front loss coefficient is around 12.6 W/m^2 K. Usually, the front loss in solar thermal collectors can be reduced by glazing. On the other hand, the back loss coefficient (U_b) is almost constant at much smaller value of 0.5 W/m^2 K such that the total overall loss coefficient(U_L) is very close to the front loss coefficient.

The corresponding heat fluxes of the PV/PCM system (q_{th}, q_{front} and q_{back}) are displayed in Fig. 14. The thermal flux absorbed by PV cell (q_{th}) is observed to be almost constant around an average value of 487 W/m^2 i.e., as high as 75% of the incident solar radiation (the remaining percentage accounts for the PV power and optical losses). This indicates that most of the total absorbed energy is converted into heat. As the front heat loss (q_{front}) is mainly depending on the front surface temperature, it has a similar shape to that of glass or PV cell temperatures (Fig. 12) with a range of 95–310 W/m^2 during the whole simulation time. This denotes about (20–64)% heat loss due to the relatively high overall front loss coefficient as demonstrated in Fig. 13. The other heat flux in Fig. 14 is

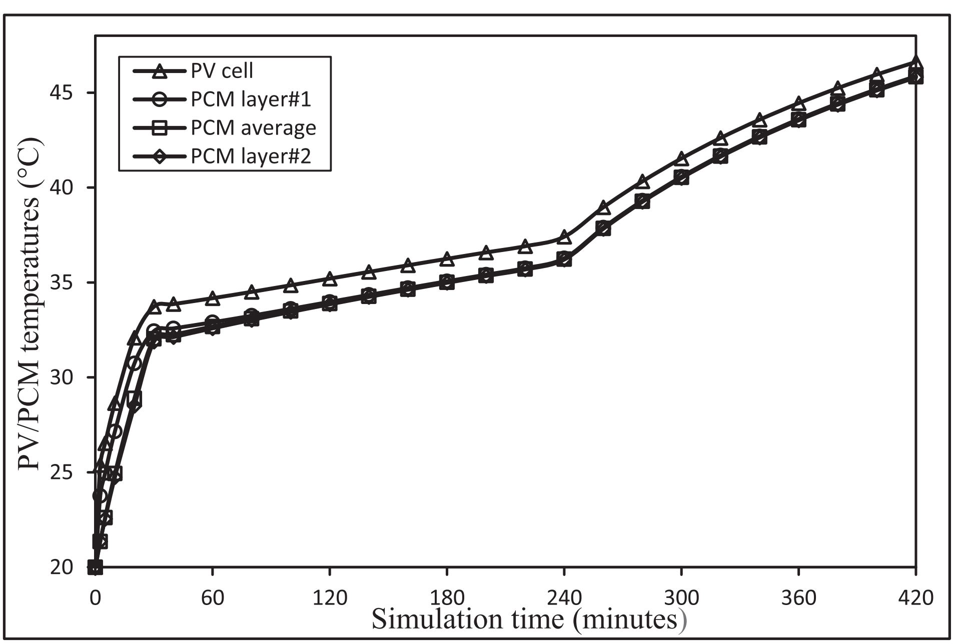

Fig. 12. Time profiles of PV cell and PCM temperatures.

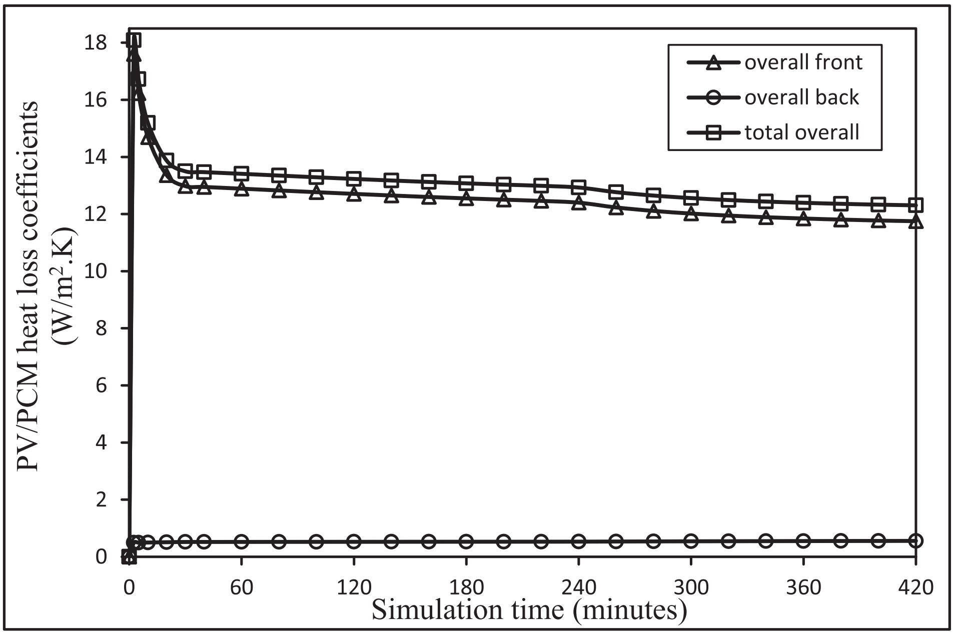

Fig. 13. Time profiles of system heat loss coefficients.

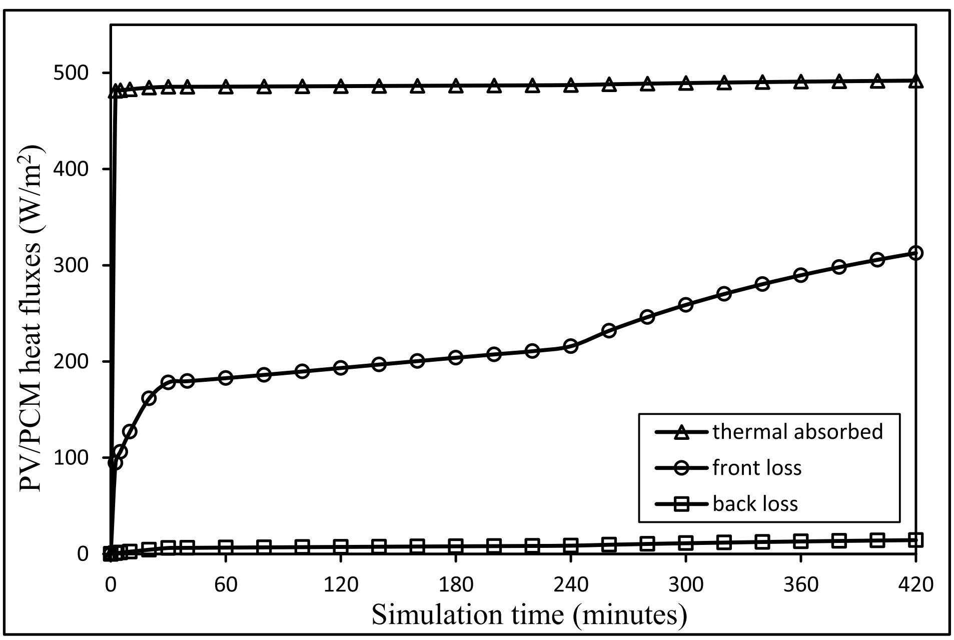

Fig. 14. Time profiles of system heat fluxes.

the back heat loss (q_{back}) which is found to be very small, less than 15 W/m^2, as usually in solar thermal collectors.

System characteristics of great significance as found by the novel PV/PCM model of the present paper are the heat transfer coefficients (U_{pcm1}, U_{pcm2} and $h_{r,pcm}$) for the thermal component (PCM) of the system as shown in Fig. 15. The heat transfer coefficient of PCM layer#1(U_{pcm1}) has three distinct regions. During the solid phase, it is very small as heat transfer process is due to conduction only. However, during the melting phase it exhibits huge upsurge from 0.75 kW/(m^2 K) to 6.7 kW/(m^2 K) due to convection mechanism induced by the liquid formation within the PCM. Then, beyond about 250 min in liquid phase, the heat transfer coefficient of layer#1 reaches saturation at 7.15 kW/(m^2 K). On the other hand, the heat transfer coefficient of PCM layer#2(U_{pcm2}) is nearly constant at much smaller value, around 0.35 kW/(m^2 K) because the thickness of layer#2 is about 20 times larger than that of layer#1. The radiative heat transfer coefficient through PCM ($h_{r,pcm}$) is negligibly small of less than 0.4 W/(m^2 K) for this case of boundary conditions.

The corresponding heat fluxes in the PCM (q_{pcm1}, q_{pcm2}, $q_{r,pcm}$ and q_u) are drawn in Fig. 16. The heat flux in PCM layer#1(q_{pcm1}) is equal to the thermal absorbed flux(q_{th}) minus the front loss(q_{front}), see Fig. 14, as the radiative flux($q_{r,pcm}$) is negligible(less than1 W/m^2) in this case study. Thus, the heat flux in layer#1 is gradually decreasing. During the melting phase, its value ranged 300–270 W/m^2 which is about 60% of the thermal absorbed flux. On the other hand, the heat flux in PCM layer#2(q_{pcm2}) is comparatively very small (7 W/m^2 during melting phase) as layer#2 has much smaller heat transfer coefficient. As a result, most portion (95%) of the heat flux in PCM layer#1 will be available for useful heat flux (q_u) as indicated by Fig. 16. Thus, the PCM is effectively

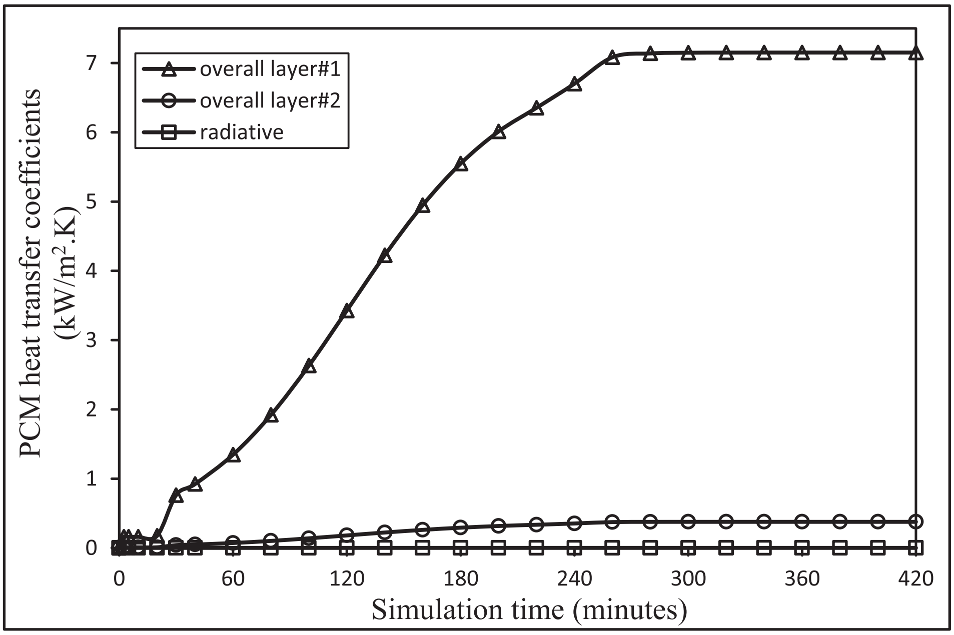

Fig. 15. Time profiles of PCM heat transfer coefficients.

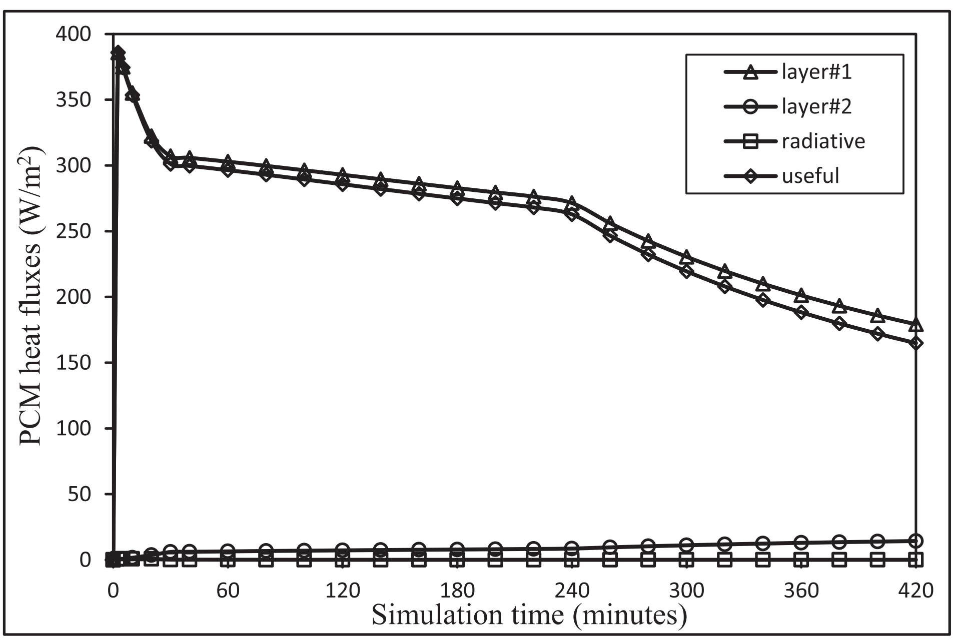

Fig. 16. Time profiles of PCM heat fluxes.

its upper layer#1 which is only 6% of the whole thermal component.

Fig. 17 shows the three different efficiencies of the system (η_{pv}, η_{th} and η_{ov}). Although PV cell temperature changed from 20 °C to 47 °C during the simulation, the corresponding PV efficiency (η_{pv}) is only varied in the range 17.3%-15.5% since the temperature coefficient of PV efficiency is relatively small (0.5%/°C). The thermal efficiency (η_{th}) clearly follows the shape of the useful heat flux (q_u) in Fig. 16 i.e., decreasing with simulation time; (60–46) % in solid phase, (46–41)% in melting phase, and (41–25)% in liquid phase. Comparatively, large drop of the thermal efficiency is observed in solid and liquid phases with only slight reduction in melting phase. As such, the total overall PV/PCM system efficiency (η_{ov}) is also decreasing as illustrated in Fig. 17.

5.2. Parametric study of system efficiencies and melting time-interval

In this section, the impact of the variation of each parameter of the five boundary conditions of the selected case study on the three system efficiencies (PV, thermal and overall) and the PCM melting time-interval will be investigated (Fig. 18**a-e**). Such parametric study can help to get better insights into the electrical and thermal behavior of the PV/PCM system, knowing that such investigations are not completely covered in the literature. It should be noted that the system efficiencies are being evaluated at the end of the melting phase as this phase is particularly used for thermal energy storage.

For the variation of the incident solar radiation (Fig. 18**a**), PV efficiency is nearly constant. In fact, a similar behavior can be noticed with the variation of other parameters (Fig. 18**b-e**) due to the small sensitivity

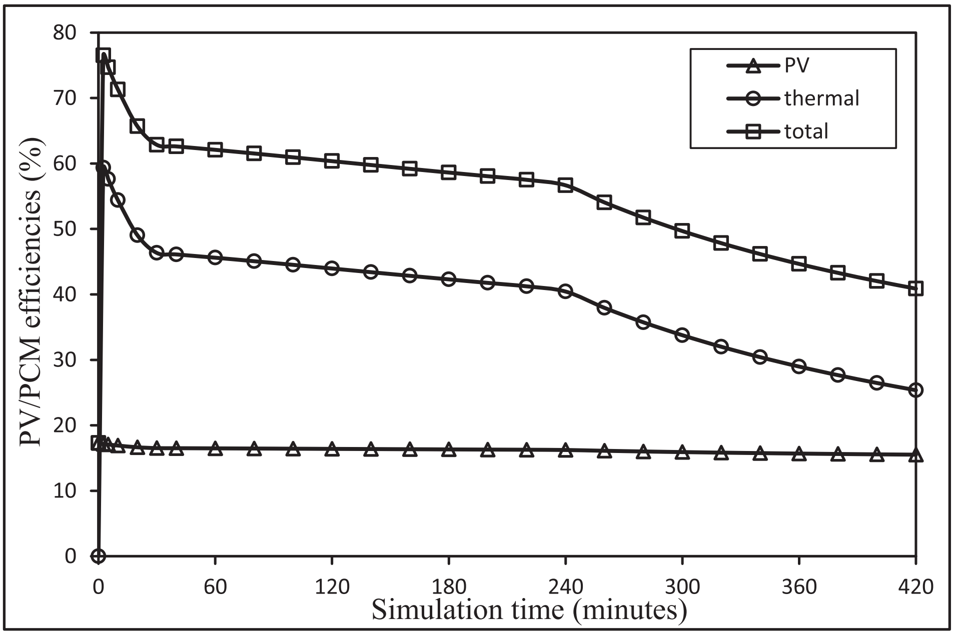

Fig. 17. Time profiles of system efficiencies.

of PV efficiency with cell temperature as previously explained. Thermal efficiency, on the other hand, increases logarithmically as available useful heat flux increases, but with a greater proportion at lower solar radiation (600 W/m^2). This is also confirmed for the overall system efficiency (Fig. 18**a),** as the PV efficiency is almost constant. In contrast, the melting time-interval is logarithmically decreasing since the heat flux in PCM layer # 1 is increasing with solar radiation, resulting in a shorter melting time-interval but with smaller variation at higher solar radiation levels (>600 W/m^2) as illustrated by Fig. 18**a**.

In Fig. 18**b**, the thermal and overall efficiencies are increasing almost linearly as the available useful heat flux is increasing due to the reduction of front thermal loss when the ambient temperature is increasing. The thermal efficiency increases by about 5 times as the ambient temperature is varied from 0 to 35 °C. Meanwhile, the melting time-interval decreases nonlinearly as heat flux increases in PCM layer # 1.

In Fig. 18**c**, the thermal and overall efficiencies are decreasing in linear proportions with the wind speed since front loss is increasing leading to reduction of the available useful heat flux. The drop in the thermal efficiency is by 55% as wind speed changed from 0 to 4 m/s. In turn, the heat flux in PCM layer#1 decreased causing the melting time-interval to be lengthened in nonlinear extend with wind speed as shown in Fig. 18**c**.

The impact of tilt angle variation is shown in Fig. 18**d** where the front loss is decreasing by the effect of the view factor. As a result, the thermal and overall efficiencies are linearly increasing while the melting time-interval is decreasing. The corresponding change in thermal efficiency and melting time-interval as tilt angle varied from 0 to 90° is 30% and −23% respectively.

For the last parameter, as the available useful heat flux is being irrespective of PCM thickness of larger than zero, the thermal and overall efficiencies are almost constant, see Fig. 18**e**. However, although the heat flux in PCM layer#1 is consequently unaffected, the melting time-interval is changing with PCM thickness by the effect of heat transfer coefficient of PCM layer#1. The latter is being reduced with PCM thickness causing longer melting time-interval at a linear rate of 8.28 min per mm of PCM thickness for the current case of boundary conditions.

6. Design charts for melting time-interval and thermal efficiency

As demonstrated in the previous section, a typical PV/PCM system could be evaluated by two performance indexes: PCM melting time-interval and the thermal efficiency according to the given parameters of boundary conditions. For that purpose, design charts are constructed in this section by applying the novel PV/PCM MATLAB program for system evaluation. Such charts can readily be used for PV/PCM design to find the corresponding performance indexes under a wide range of boundary conditions. The design charts of Fig. 19**a-c** are applicable for incident solar radiation of 300–1200 W/m^2, PCM thickness of 5–100 mm, and ambient temperatures of 10 °C, 20 °C, and 30 °C. For convenience, the other two parameters of the boundary conditions are taken to be fixed at 90° tilt angle and zero wind speed. Thus, Fig. 19**a-c** will give minimum melting time-intervals and maximum thermal efficiencies in respect to other values of tilt angle and wind speed. It can be observed from the design figures that for given solar radiation and ambient temperature, the system exhibits constant thermal efficiency at any PCM thickness, while the melting time-interval is in linear proportion to the PCM thickness, however with larger rates at lower solar radiation. At ambient temperature 20 °C, the corresponding rate is 3.48 and 20.85 min/mm at 1000 and 300 W/m^2 respectively as can be found from Fig. 19**b**. Furthermore, a higher solar radiation results in a higher thermal efficiency, but only to a logarithmic extent, as described in the previous section. Comparing the three design charts, a PV/PCM system at a higher ambient temperature is found to attain a greater thermal efficiency. However, the improvement in the thermal efficiency of the system due to effect of both solar radiation and ambient temperature is such that as the solar radiation varies from 300 to 1000 W/m^2, the corresponding improvement is 229.8%, 75.4%, and 23.2% at ambient temperature 10, 20, and 30 °C respectively. On the other hand, for a higher ambient temperature the melting time-interval per mm of PCM thickness would be of a lower rate at a given incident solar radiation. At 400 W/m^2, this rate is 18.0, 12.2, and 9.1 min/mm at ambient temperature 10, 20, and 30 °C respectively as can be obtained from the three design charts. For PV/PCM design at a given incident solar radiation, ambient temperature, wind speed, and tilt angle, the thermal efficiency of the system is being determined accordingly. Then, the PCM thickness can be specified such as to get the required melting time-interval as the thermal efficiency is unaffected by the thickness of the PCM.

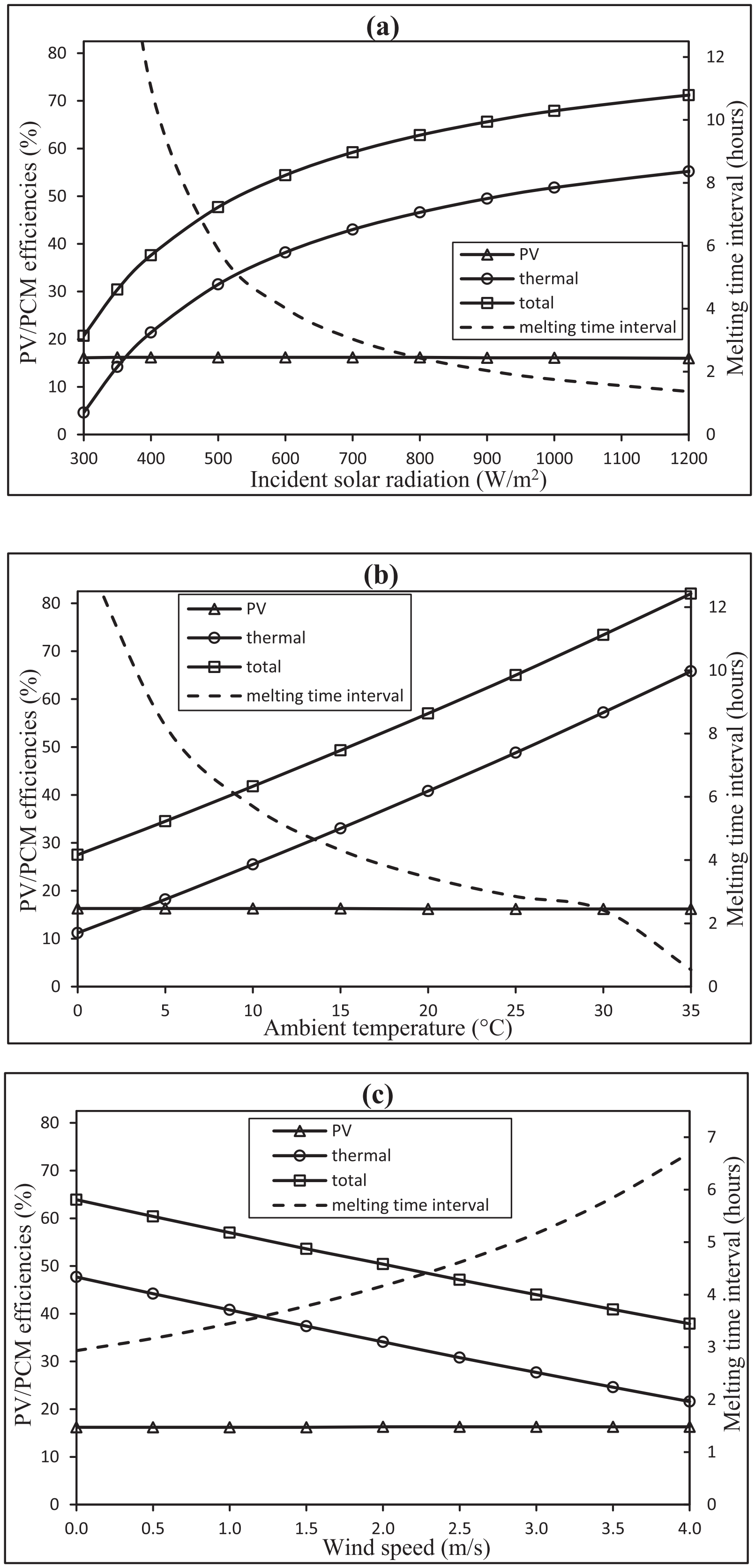

Fig. 18. Variation of system efficiencies and melting time-interval with (a) incident solar radiation (b) ambient temperature (c) wind speed (d) tilt angle and (e) PCM thickness.

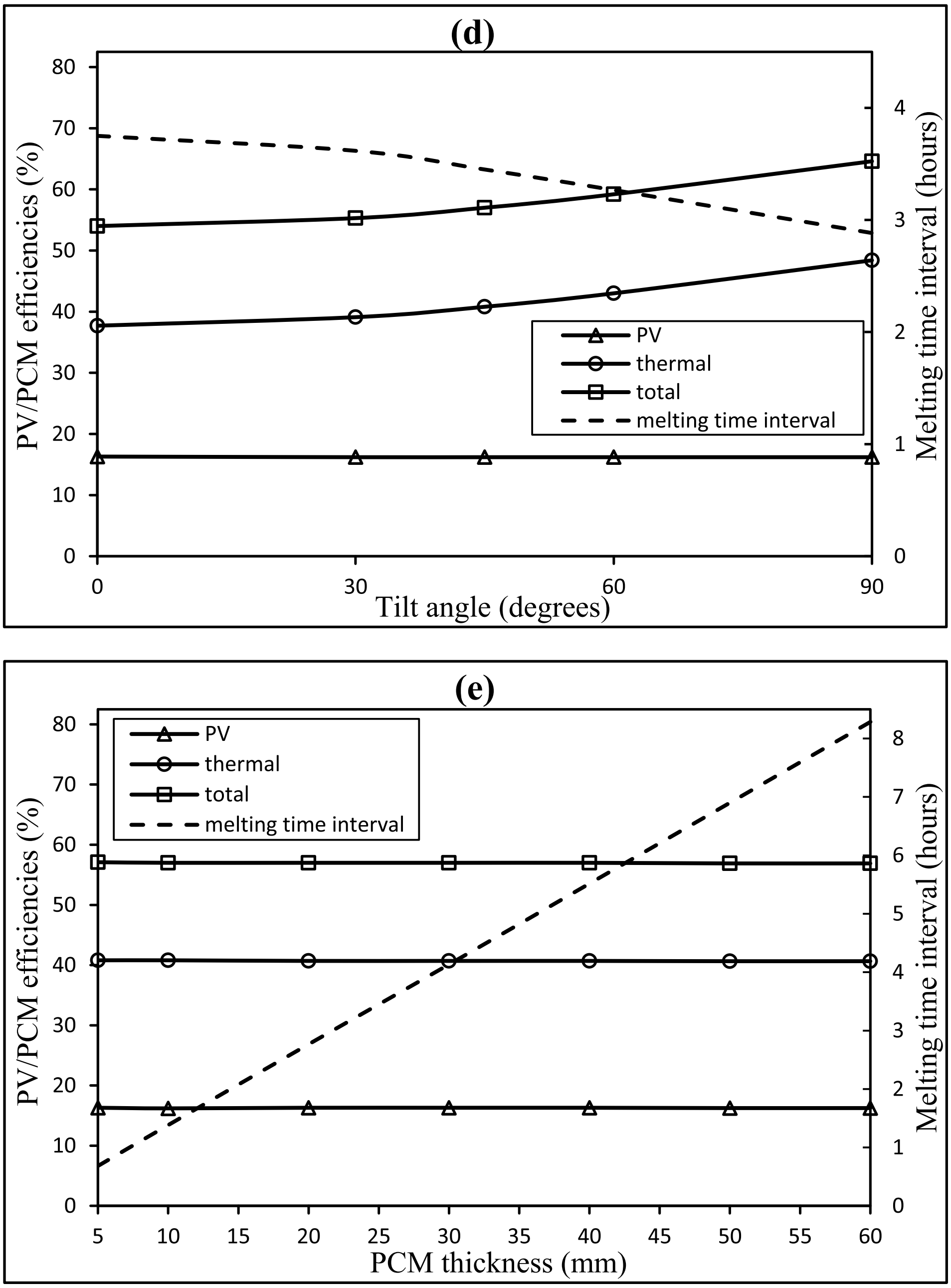

Fig. 18. (*continued*).

7. Conclusions

A novel mathematical model was developed in this paper for a typical PV/PCM system with a better compromise between accuracy, computation time, and simplicity in comparison to the commonly used CFD model. The new model was first validated against both CFD numerical and experimental results, and then a case study was adopted to explore the various characteristics and parametric performance of the system. Finally, design charts for the two system indexes: PCM melting time-interval and thermal efficiency, were constructed using the novel model such that it can readily be applied under a wide range of boundary conditions. Based on this research, the following conclusions could be drawn:

1. The CFD numerical validation indicated good accuracy of the novel model with a normalized root mean square error of 2.5% and a linear coefficient of correlation of 0.987. The experimental validation exhibited even better accuracy with a corresponding error of 1.4% and a correlation coefficient of 0.999. Thus, the validity of the proposed model for solving PCM-based systems is confirmed as a new numerical method.
2. The novel PV/PCM model was developed as a MATLAB model, employing simple numerical functions without mesh patterns. The developed model is about 4000 times faster than the corresponding CFD model. Consequently, comprehensive investigations of a wide range of PV/PCM system variables were carried out in this research with >5000 simulations including model adjustment and validation, time profiles of system characteristics, parametric studies, and design charts.
3. The position of the average PCM temperature is vital for the mathematical modelling of the system as it divides the PCM into two layers: upper and lower. This position was found to be at about

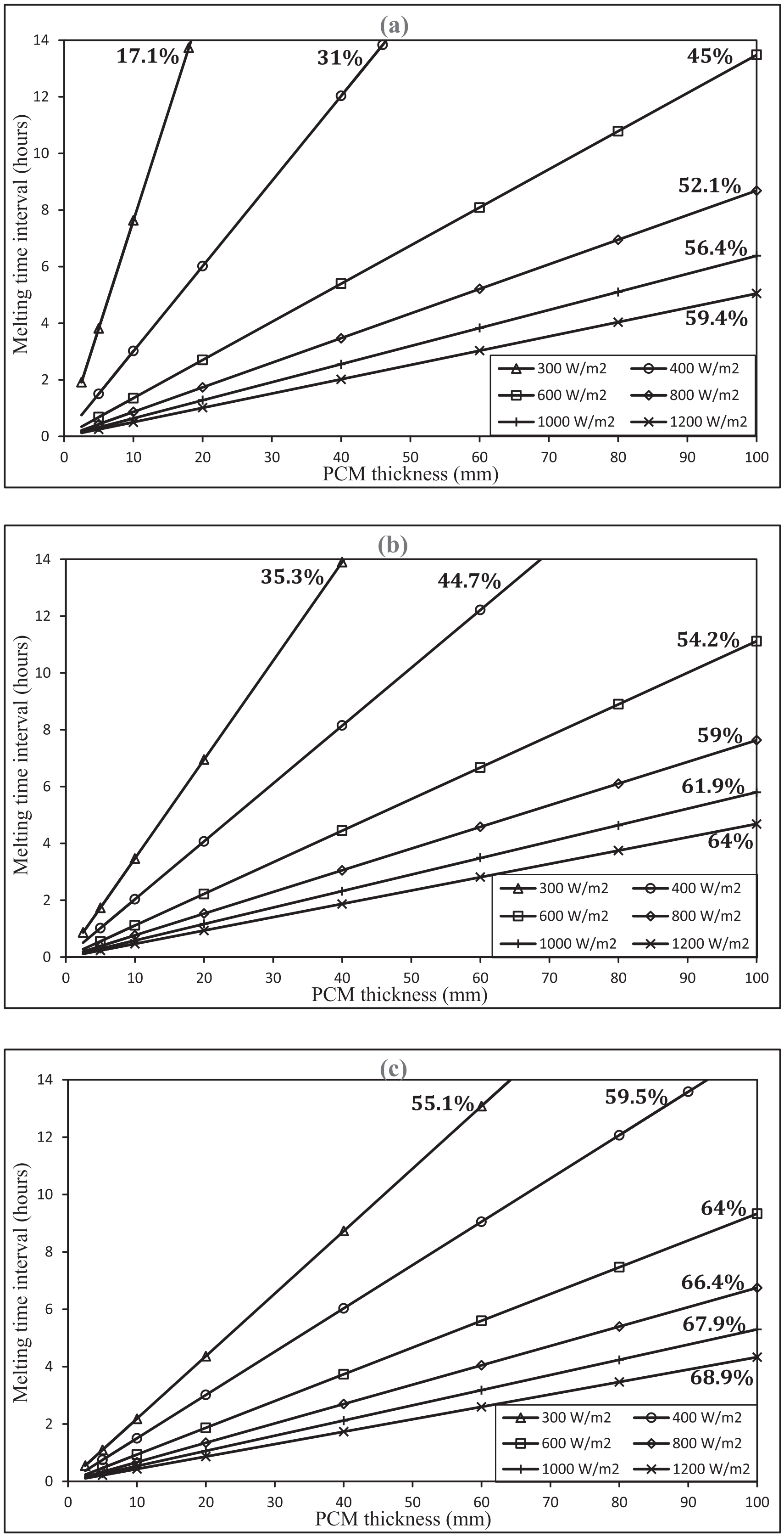

Fig. 19. Design charts for a typical PV/PCM system with 90° tilt angle and zero wind speed at ambient temperatures (a) 10 °C (b) 20 °C and (c) 30 °C.

(5–6)% of the total PCM thickness as measured from the upper side of the PCM, with an average temperature close to that of the lower side. Thus, the PCM is more effective at its upper layer, which governs the melting time-interval and the thermal efficiency of the system as its heat transfer coefficient is about 20 times higher than that of the lower layer. The heat transfer coefficient of the PCM's upper layer could be as large as 6.7 $kW/(m^2 K)$ at the end of the melting phase, as obtained in the case study, due to the convection mechanism induced by the liquid formation within the PCM layers.

4. Relatively high front loss coefficient is observed in the PV/PCM system due to the considerable effect of the large deviation of sky temperature from ambient (>10 °C), as also noticed in solar thermal collectors. It is around 12.6 $W/(m^2 K)$ during the melting phase of the PCM.
5. Because of the low sensitivity of PV efficiency to cell temperature variation, PV efficiency varied with a limited range of 17.3% to 15.5% as the PV cell temperature varied from 20 °C to 47 °C during the case study simulation. As the front loss of the system increases over the solid, melting, and liquid phases of the PCM, the corresponding thermal efficiency is (60–25%). When thermal energy is stored in PCM during the melting phase, the thermal efficiency is nearly constant at (46–41%).
6. The impact of ambient temperature, as illustrated by the parametric study, is such that the thermal efficiency of the system increases almost linearly by 5 times as the ambient temperature varies from 0 to 35 °C. While the melting time-interval is shortening nonlinearly due to the increase in heat flux in the upper PCM layer, For the wind speed, the thermal efficiency drops by 55% as the wind speed varied from 0 to 4 m/s since the front loss was increasing, causing the melting time-interval to be longer on a nonlinear scale. The effect of tilt angle variation from 0 to 90° is such that the corresponding change in thermal efficiency and melting time-interval is 30% and 23%, respectively.
7. According to the performance analysis and design charts, the thermal efficiency of the system is logarithmically proportional to the incident solar radiation. While the melting time-interval is also logarithmic, but of inverse proportion. Both changed at larger rates at lower solar radiation levels. For PCM thickness, the thermal efficiency is unaffected, whereas the melting time-interval varies at a linear rate with the effect of the heat transfer coefficient of the upper PCM layer. For the case study, the rate of change is 8.28 min per mm of PCM thickness. At a higher ambient temperature, the rate of change for the melting time-interval with PCM thickness is smaller due to the reduction in front loss.

CRediT authorship contribution statement

Hussein M. Taqi Al-Najjar: Conceptualization, Methodology, Software, Investigation, Writing – original draft. **Jasim M. Mahdi:** Data curation, Investigation, Validation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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